Information Security

SPRING 2023

Lecture 7: Vigenere Cipher

Vigenère Cipher

- Vigenère Cipher is a method of encrypting alphabetic text. It uses a simple form of polyalphabetic substitution.
- A polyalphabetic cipher is any cipher based on substitution, using multiple substitution alphabets.
- The encryption of the original text is done using the Vigenère square or Vigenère table.
- The table consists of the alphabets written out 26 times in different rows, each alphabet shifted cyclically to the left compared to the previous alphabet. (shown in next slide).

Vigenère Cipher

- A keyword is used for encryption process.

For example:

Plain Text= “SACK GAUL SPARE NO ONE”.

Keyword: ITALY

- For generating key, the given keyword is repeated in a circular manner until it matches the length of the plain text.

I T A L Y I T A L Y I T A L Y I T A

S A C K G A U L S P A R E N O O N E

Vigenère Cipher

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

Vigenère Cipher

- **Rules of encryption is as follows:**
- **Step-1:** Create Vigenère table as per your given keyword. (If Vigenère table is not given).
  - Given keyword arrange in alphabetical order then create table as per given below.

For Example,

Keyword: MEC

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L

Vigenère Cipher

- **Rules of encryption is as follows:**
- **Step-2:** To perform the substitution, start with the first combination of keyword and message letters IS.
  - Use the keyword letter to locate the column, and the message letter to find the row, and then look for the letter at their intersection.
  - Thus, for column “I” and row “S,” you will find the ciphertext letter “A.”
  - After you follow this procedure for each of the letters in the message, you will produce the encrypted ciphertext = ATCVEINLDNI KEYMWGE.

Vigenère Cipher

- **Rules of encryption is as follows:**
- **Step-3:** In Keyword length is lesser than given plain text then repeat keyword alphabets until it matches the length of plain text.
- **For example,**
- **Plain Text: I AM PAKISTANI, Keyword: MEC**

Keyword is repeated →

Key:	**M**	**E**	**C**	**M**	**E**	**C**	**M**	**E**	**C**	**M**	**E**	**C**
Plain Text:	I	A	M	P	A	K	I	S	T	A	N	I

Vigenère Cipher

- **Step-4:** The first letter of the plaintext, ***"I"*** is paired with ***"M"***, the first letter of the key. So, use row ***"M"*** and column ***"I"*** of the Vigenère table, it produces cipher text ***"U"*** and so on.
- **Plain Text: I AM PAKISTANI, Keyword: MEC**

Key:	**M**	**E**	**C**	**M**	**E**	**C**	**M**	**E**	**C**	**M**	**E**	**C**
Plain Text:	I	A	M	P	A	K	I	S	T	A	N	I
Cipher Text:	U	E	O	B	E	M	U	W	V	M	R	K

Class Task

- **Question:** Find out cipher text of below plain text using "Vigenere Cipher".
  - **Plain Text:** DO NOT WASTE YOUR TIME
  - **Keyword:** CAN

Summary of today's lecture

- We discussed two types of substitution cipher playfair cipher and vigenere cipher.